

Functional sequences, series and spaces – 1

Sequences of functions and uniform convergence

13 сентября 2026 года

Theoretical minimum.

Definition and criterion. $f_n \rightarrow f$ uniformly on X means $\sup_{x \in X} |f_n(x) - f(x)| \rightarrow 0$; equivalently the Cauchy condition $\sup_x |f_n(x) - f_m(x)| \rightarrow 0$ holds. Pointwise convergence is the statement with the supremum replaced by “for each x ”, and the whole subject consists of understanding which properties survive the difference.

Theorem 1 (what uniform convergence preserves). A uniform limit of continuous functions is continuous, a uniform limit of Riemann integrable functions on $[a, b]$ is integrable and $\int f_n \rightarrow \int f$. Differentiability is *not* preserved: what is true is that if $f_n \in C^1$, $f_n(x_0)$ converges and $f'_n \rightarrow g$ uniformly, then f_n converges uniformly to a function f with $f' = g$. The hypothesis is on the derivatives, not on the functions.

Theorem 2 (Dini and Pólya). If continuous f_n decrease pointwise to a continuous limit on a compact set, the convergence is uniform (Dini, proved in Continuous functions – 7); if monotone functions converge pointwise to a continuous limit on $[a, b]$, the convergence is again uniform (Pólya). Both are statements that monotonicity in one of the two variables upgrades pointwise to uniform.

Theorem 3 (Arzelà–Ascoli). A family $\mathcal{F} \subset C(K)$ on a compact K is precompact in the uniform norm if and only if it is uniformly bounded and equicontinuous. Neither hypothesis can be dropped, and the theorem is the only general source of convergent subsequences in a space of functions.

Fact 1 (the standard zoo). x^n on $[0, 1]$: pointwise, not uniformly. nxe^{-nx^2} :

pointwise to 0, not uniformly, integrals converge to a non-zero limit. $\frac{\sin nx}{\sqrt{n}}$: uniformly to 0 while the derivatives blow up. $\frac{x}{1+n^2x^2}$: uniformly to 0, derivatives converge pointwise to a discontinuous function. Every counterexample in this subject is one of these four in disguise.

1. Prove the supremum criterion and the Cauchy criterion for uniform convergence, and prove Theorem 1 in all three parts. Then examine the four examples of Fact 1 and say in each case exactly which hypothesis of Theorem 1 fails.
2. Prove the Arzelà–Ascoli theorem in both directions, and show by examples that uniform boundedness without equicontinuity and equicontinuity without uniform boundedness both fail to give precompactness.
3. (Kapačëв 1.83) A sequence of uniformly continuous functions $f_n: \mathbb{R} \rightarrow \mathbb{R}$ converges uniformly to $f: \mathbb{R} \rightarrow \mathbb{R}$. Does it follow that f is uniformly continuous? What changes if the convergence is only pointwise?
4. (Kapačëв 1.84) Let (f_n) be a uniformly bounded sequence of monotone functions on $[a, b]$. Prove that some subsequence converges pointwise (Helly’s selection theorem), and decide whether the limit must be monotone.
5. (ISTCiM 2023) Let $f_n: [0, 1] \rightarrow \mathbb{R}$ be given by $f_n(x) = \frac{1}{n+1} \left(1 + \frac{x}{1}\right) \left(1 + \frac{x}{2}\right) \cdots \left(1 + \frac{x}{n}\right)$. Show that (f_n) converges uniformly on $[0, x]$ for every $0 < x < 1$, and find the pointwise limit.